

SmartSoft: The Framework For Robotic Systems

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Artificial intelligence, mechatronics, robotics—these are fields in demand in today's world, thanks to the dreams of living luxurious lives, having automatic facilities

all around and having to put in less manual effort. But one fact remains; although the process of developing such a solution has eased in comparison, complexity still stays intact.

In this article we introduce a framework for developing component based robotic systems. To put it in a sentence, SmartSoft is an open source integrated model-driven development environment for robotics software. Built with a vision of creating a business ecosystem for robotics software, SmartSoft takes the service-oriented software component model and model-driven toolchain approach. The software is presently under Service Robotics Research Center of University of Applied Sciences Ulm, Germany, with the team headed by Prof. Dr Christian Schlegel.

What SmartSoft provides

The software aims at assisting the researcher, the robotics professional, the system integrator, the application domain expert and the end user, in realising their ideas. The package offers model-driven tools for component developers, system integrators and professional end users. The designer can make use of component selection guides, and automatic generation of component hulls for wrapping algorithms and ensuring system-level composability.

Further, the kit provides component repositories for navigation, vision, human-machine interface, task coordination, manipulation and a lot more that go into the simplest of robotic designs.

The SmartSoft approach

The factor that serves as a differentiator for SmartSoft is its approach towards robotics. Adopting a component based view, SmartSoft looks at each component model in terms of communication patterns, which

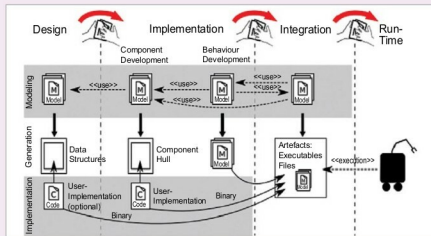


Fig. 1: Internal realisation of SmartMDSOToolchain. Figure shows artefacts of the toolchain (models and code), their relationships and their handover (red arrows) through the workflow

For curious minds

- ▶ SmartSoft assists the component developer, the application builder and the end user in building and using distributed components in such a way that semantics of the interface of components is predefined by communication patterns, irrespective of where these are applied.
- ▶ At the moment, the most prominent implementations of SmartSoft are based on ACE/TAO named CORBA/SmartSoft and on adaptive communication environment (ACE) named ACE/SmartSoft.
- ▶ CORBA/SmartSoft runs on Linux out-of-the-box. Others use it on Windows. ACE/SmartSoft runs on Linux and Windows out-of-the-box. Since it depends only on ACE and relies on its operating system abstractions, it requires only minor efforts to get it into operation on all operating systems supported by ACE.
- ▶ Other implementations of SmartSoft are:
 - SmartSLAM is an open source C++ library for simultaneous localisation and mapping algorithms in robotics based on probabilistic methods
 - SmartSoft ROS gateway
 - SmartSoft-Morse simulator
- ▶ SmartSoft Component Model is underpinned by a unified modelling language (UML) profile. Thus, SmartSoft approach is accessible via model-driven software development approach (MDSO).
- ▶ In order to reuse existing components and algorithms on your robot, install an implementation of SmartSoft (ACE/SmartSoft installation guide). However, you do not need to install the eclipse based SmartMDSOToolchain.
- ▶ In order to start developing your own components and deploying these on your robot, you will additionally need to install SmartMDSOToolchain.

—Source: SmartSoft website, www.servicerobotik-ulm.de/drupal